

Better relationships with parents make children less promiscuous

Evidence for developmental calibration of sexual strategies from a genetically informed adoption study of 617 families

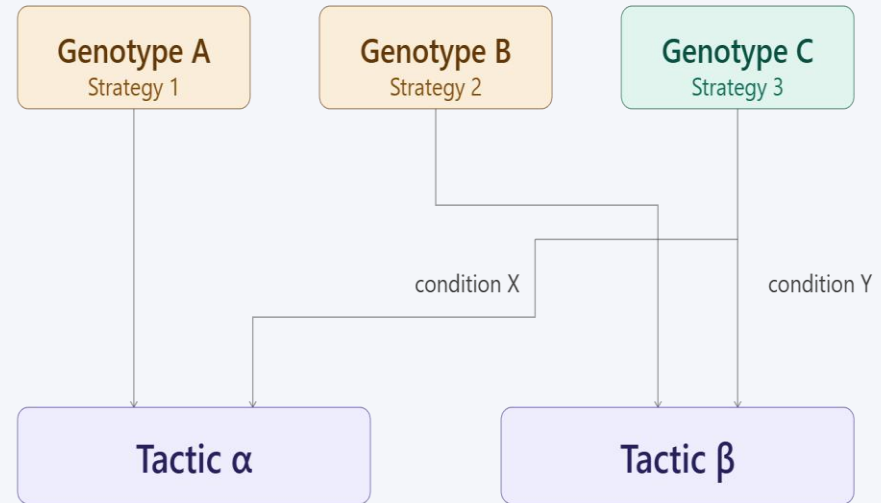
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Evolutionary Strategies and Tactics

What are strategies and tactics?

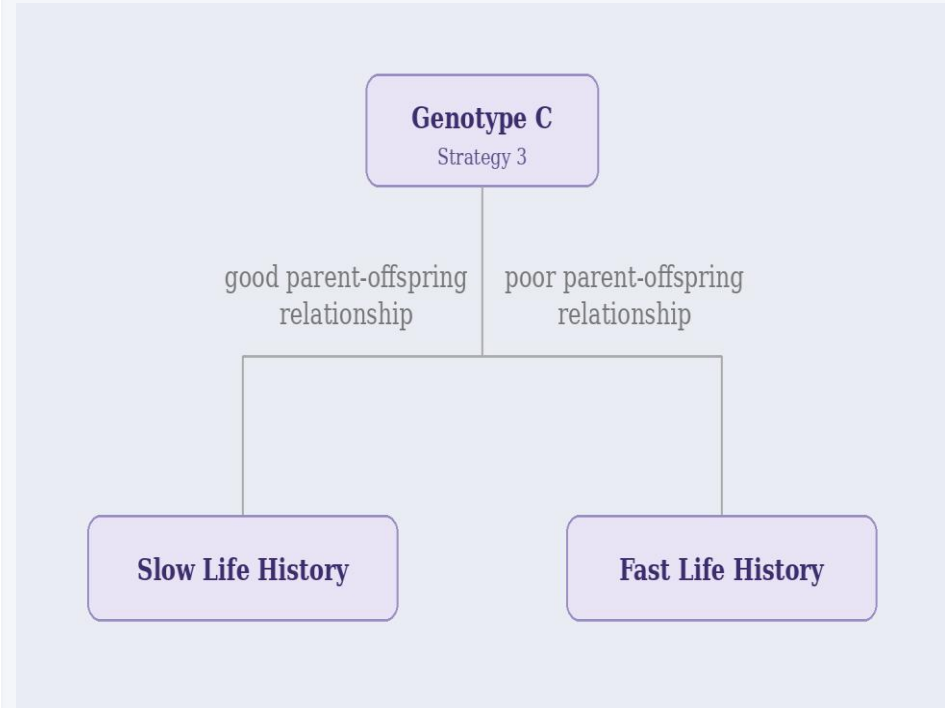
- Strategies = ‘evolved set of rules for expressing tactics’ (Dominey, 1984)
- Tactics = behavioral phenotype
- Tactic variation $\neq$ strategy variation



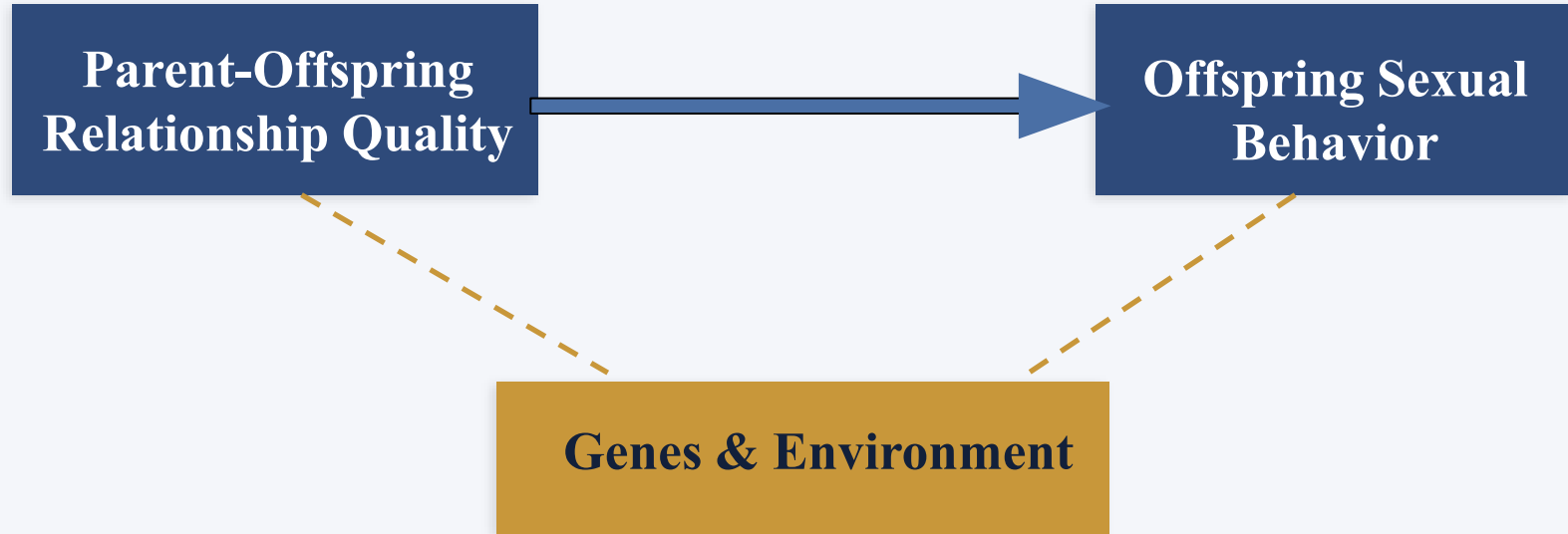
Evo-Devo:

Adult Phenotype Dependent on Early Environment

- Early environmental cues calibrate offspring reproductive tactics (Ellis et al., 2009; 2022)
- Predictive Adaptive Responses (Nettle et al., 2013)
- Parent-offspring relationship quality as a calibrating signal (Belsky et al., 1991)



The Causal Model



Study Design

Minnesota Sibling Interaction and Behavior Study Dataset

- 617 two-offspring families (1,234 children)
- 55% female
- Time 1: Parent-offspring relationship quality (M age = 17.1)
- Time 2: Offspring sexual behavior (M age = 20.6)

Family Composition

 $N = 285$ (46%)

Two adopted offspring
(0% shared genes)

 $N = 124$ (20%)

One adopted + one non-adopted
(0% shared genes)

 $N = 208$ (34%)

Two non-adopted offspring
(50% shared genes)

Parent-Offspring Relationship Quality

Composite of Four Dimensions

Conflict (12 items)

"My parent often criticizes me"

Involvement (12 items)

"I talk about my concerns with my parent"

Parent Regard (5 items)

"I know my parent loves me"

Offspring Regard (8 items)

"I respect my parent"

Multiple Reporters

Parent Reported

Average of separate
mother and father
reports ($r = .31$)

Offspring Reported

Average of offsprings
perception of
relationship with mother
and father ($r = .80$)

Sexual Behavior

Sexual Behavior Inventory (SBI; 12 items)

Casual oral sex

Committed oral sex

Casual vaginal/anal sex

Committed vaginal/anal sex

E.g., How many different committed romantic partners have you had oral sex with in the last 12 months?

Three questions within each category: ever engaged (yes/no), number of partners in last year, lifetime numbers of partners

Single factor across all 12 items was reliable and homogenous ($u = .85$, $\omega_h = .73$, $\alpha = .93$).

Analytic Strategy: Three Lines of Evidence

1

Biometric ACE Modeling

2

Passive *r*GE Test

3

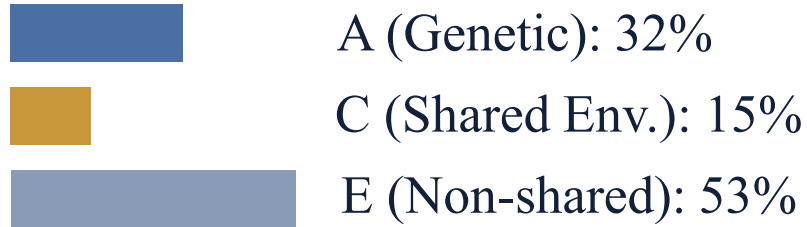
Within-Family Sibling Comparison

Prospective design reduces (but cannot eliminate) reverse causality

Shared Environment Effects

15% of the variance in offspring sexual behavior reflected shared environment effects

Best-Fitting ACE Model

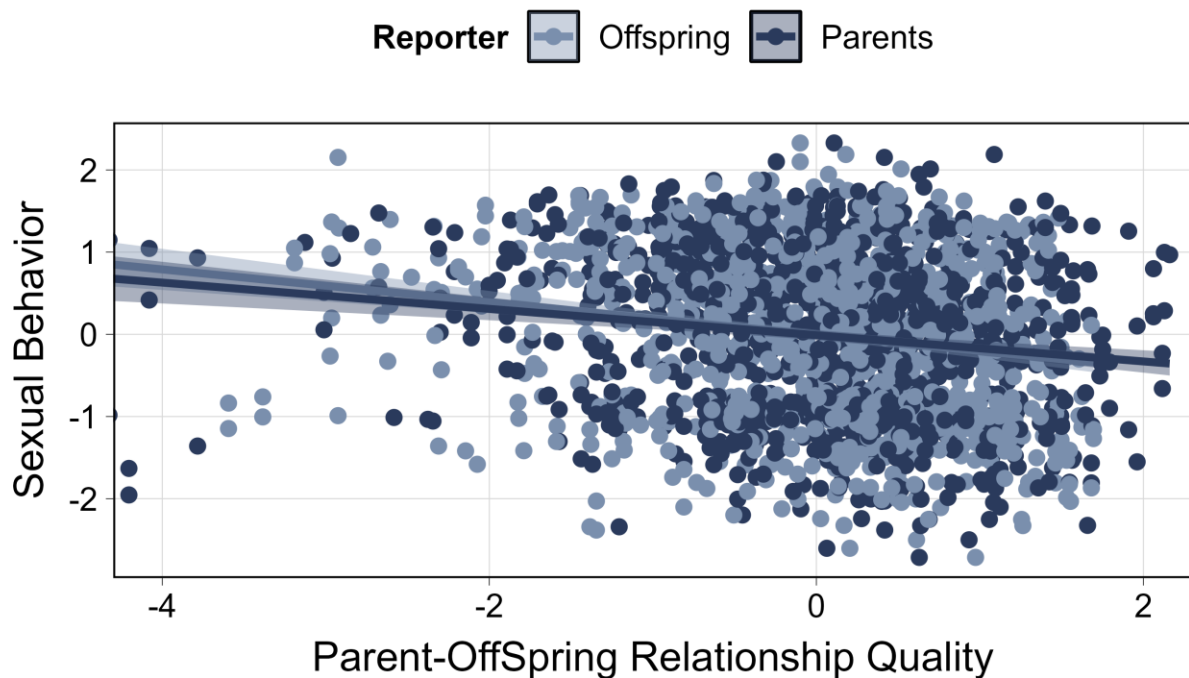


Other twin and adoptee studies also find shared environmental effects on sexual behavior

(e.g., Bricker et al., 2006; Harden & Mendle, 2011; Samek et al., 2014; Zietsch et al., 2010)

Between-Family Associations

Families with higher average parent-offspring relationship quality have offspring with lower sexual behavior



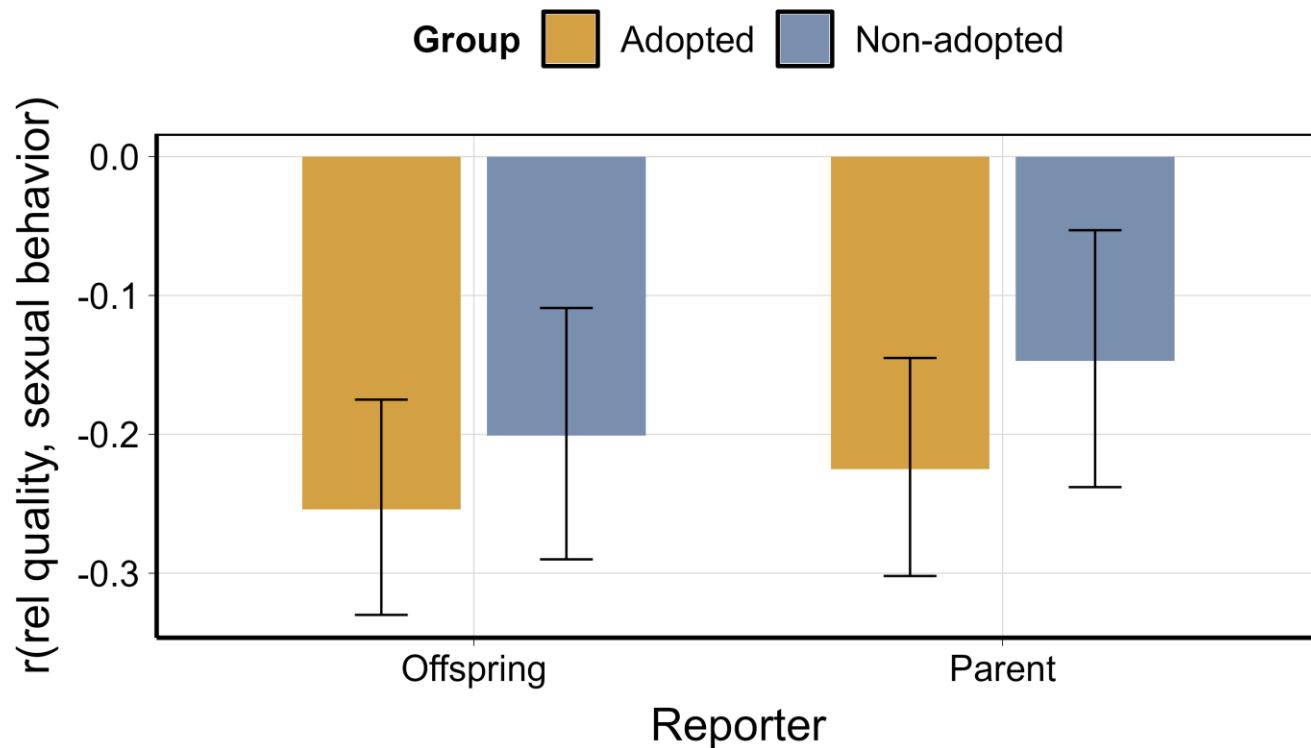
Offspring reports:

$$\beta = -0.20 [-0.27, -0.13],$$
$$p < .001$$

Parent reports:

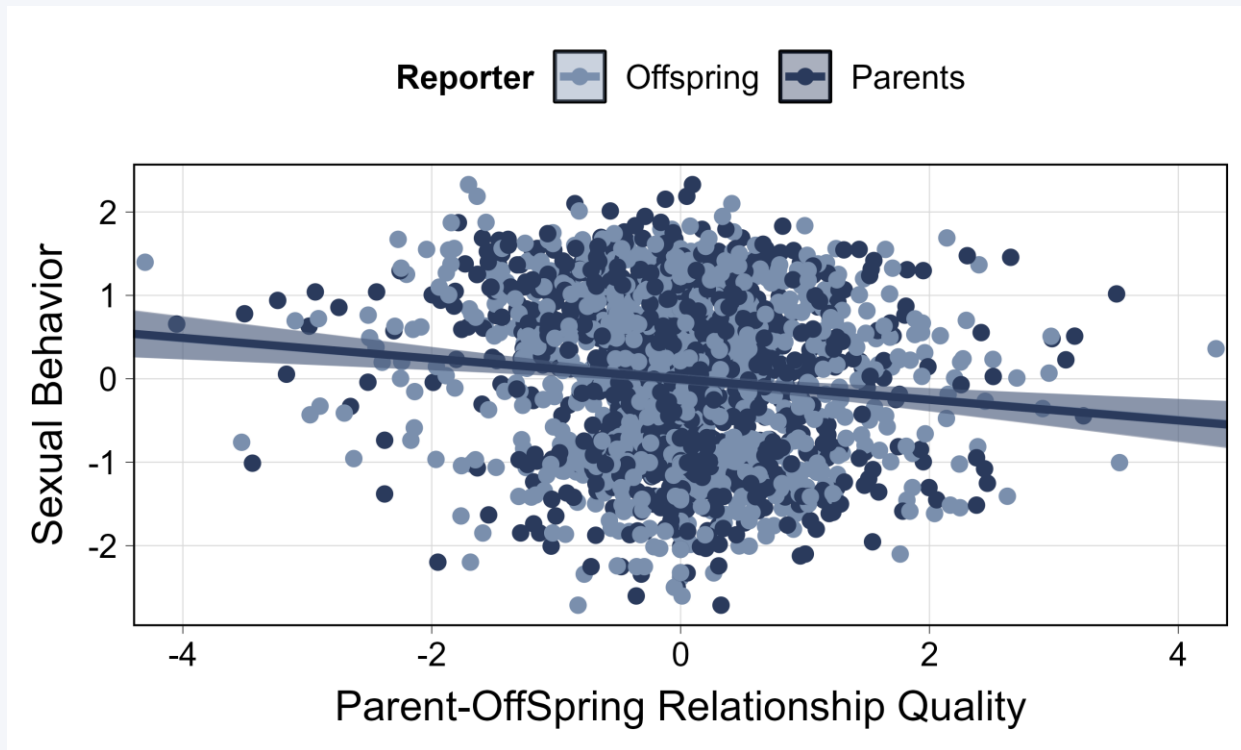
$$\beta = -0.14 [-0.21, -0.08],$$
$$p < .001$$

No Passive r GE Confounding Detected



Within-Family Associations

Siblings who have a higher quality relationship with their parents have lower sexual behavior relative to their sibling.



Offspring:

$\beta = -0.12$ [-0.18, -0.06],
 $p < .001$

Parents:

$\beta = -0.12$ [-0.17, -0.06],
 $p < .001$

Robustness Analyses

Results were remarkably robust to different analysis decisions

No offspring sex moderation detected

Mom and dad reports yielded consistent effects when tested separately

Consistent effects across casual and committed partner dimensions

What is the Mechanism?

Fewer sexual partners can arise through very different pathways

Pathway 1

↓ Uncommitted Sex

Better parenting reduces propensity for casual sexual encounters — fewer short-term mating attempts overall

Pathway 2

↑ Relationship Stability

Better parenting promotes stable committed relationships — offspring cycle through fewer partners because relationships last longer

Pathway 3

Something Else?

Better mate choice?

Reduced mating effort?

Sibling dynamics?

Summary

- Better parent-offspring relationships prospectively predict fewer sexual partners
- This association is not explained by passive rGE (genetic confound)
- Within-family associations are consistent with causal environmental effect
- Results are robust to analysis decisions
- The operative mechanism remains unknown
- Risky tests of developmental calibration require establishing the reliability and validity of parent-offspring relationship quality as a cue for adult environments